

CI/CD Report — HW06 API Testing

Student ID: 23127152
Pipeline: GitHub Actions + Newman
Date: 2026-08-19

1. Pipeline Configuration

Workflow file: [.github/workflows/hw06-api-tests.yml](#)

Trigger: push / pull_request on hw6/23127152 (paths: 23127152-hw6/**, backend/**, workflow file) + workflow_dispatch.

Job steps (summary):

- 1. Checkout repository
- 2. Setup Node.js 20
- 3. npm install in backend/
- 4. Start SUT: node server.js in background (127.0.0.1:3000)
- 5. Wait until GET /api/products succeeds
- 6. Install newman + newman-reporter-htmlextra
- 7. Run folder **CI Smoke – Must Pass** (green happy-path suite; full SEC negative suite stays local/Phase 1–3)
- 8. Upload HTML artifact newman-ci-smoke-report
- 9. Every request still carries X-Student-Id: 23127152 via collection pre-request

```
# Key Newman invocation (from workflow)
newman run postman/EShop-HW06.postman_collection.json \
  -e postman/EShop-HW06.postman_environment.json \
  --folder "CI Smoke – Must Pass" \
  --env-var "baseUrl=http://127.0.0.1:3000" \
  --env-var "studentId=23127152" \
  -r cli,htmlextra \
  --reporter-htmlextra-export postman/reports/newman-ci-smoke.html
```

ciExpectStatus is read from the Postman environment (default **200**). The intentional-fail sample set it to **201** so CI-01 expects the wrong status.

2. Sample Run — All Passing

Item	Value
Commit SHA	8b44b66
Commit message	hw06/cicd-github-actions
Pipeline URL	https://github.com/yuran1811/hcmus-sw-testing--eshop-sut/actions/runs/32269053058
Result	✅ success — all CI Smoke assertions passed
Screenshot	cicd/screenshots/run-all-pass.png (Actions UI) · log: run-all-pass-log.txt

CI Smoke: 4 requests · 9 assertions · 0 failed. Backend booted on Actions runner; Newman talked to real SUT on 127.0.0.1:3000.

3. Sample Run — One Test Failing

Item	Value
Commit SHA	2fe51ee
Commit message	hw06/cicd-intentional-fail-v2
Pipeline URL	https://github.com/yuran1811/hcmus-sw-testing--eshop-sut/actions/runs/32269286455
Result	 failure
Failed test	CI-01 status — expected 201 , got 200
Screenshot	cicd/screenshots/run-one-fail.png (Actions UI) · log: run-one-fail-log.txt

Why intentional: demonstrate a red pipeline by changing only the expected status for CI-01 (`ciExpectStatus=201` / `expect 201`), not by breaking the SUT. Job still started backend successfully; Newman exit code 1 failed the workflow.

4. Observations

- Separating **CI Smoke** (always-green contract) from the full security suite (expected fails documenting bugs) keeps the pipeline useful as a regression gate.
 - First intentional-fail attempt (86cd4a6) stayed green because the workflow hard-coded `--env-var ciExpectStatus=200`, overriding the env file — fixed in 2fe51ee by removing that override.
 - After the red sample, suite was restored to `ciExpectStatus=200` so subsequent pushes are green again.
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5. Related links

Item	URL
Workflow runs (branch)	https://github.com/yuran1811/hcmus-sw-testing--eshop-sut/actions/workflows/hw06-api-tests.yml
Pass run	https://github.com/yuran1811/hcmus-sw-testing--eshop-sut/actions/runs/32269053058
Fail run	https://github.com/yuran1811/hcmus-sw-testing--eshop-sut/actions/runs/32269286455